

Term paper

“Practical Studies Industry 4.0”

Hof University of Applied Sciences

Software Engineering for Industrial Applications

Submitted to
Prof. Dr. Heike Markus
Alfons-Goppel-Platz 1
95028 Hof
Germany

Submitted by
Emil Shkurti
Ludwigstraße 18
95028 Hof
Germany

1. Introduction

1.1. Context and Problem Statement

Modern automotive workshops increasingly rely on digital service manuals and technical knowledge bases to manage the growing complexity of vehicle systems. These manuals contain hierarchical and multimodal content, including procedural instructions, schematics, tables, and annotated images, which technicians must interpret accurately during maintenance, diagnostics, and repair tasks. As vehicles incorporate more advanced electronic control units and interconnected subsystems, manual lookup in such documentation becomes both time-consuming and prone to error. This challenge is amplified for service procedures that depend heavily on visual interpretation of diagrams or the integration of information spread across multiple hierarchical sections.

1.2. Project Scope and Proposed Solution

To explore a scalable technological solution to these challenges, this project considers a fictitious software provider, AutoTech Solutions GmbH, which develops knowledge-assistance systems for authorized automotive workshops. The company operates on the assumption that workshops have licensed access to OEM or OEM-approved third-party digital manuals. In many real-world settings, these manuals are provided in structured or semi-structured HTML formats, accompanied by technical images and diagrams. This project mirrors that setting by using an HTML-based vehicle service manual containing technical diagrams, procedural instructions, and specification tables typical of automotive repair manuals.

The goal of the project is to investigate whether multimodal Retrieval-Augmented Generation (RAG) can support technicians in extracting precise information from such densely structured documentation. Instead of relying on text-only retrieval, the system integrates textual hierarchy information with relevant visual material during the reasoning stage, allowing large language models (LLMs) to ground their responses in diagrams, tables, and procedural images. This capability aims to reduce lookup time and improve interpretation accuracy without replacing human decision-making.

1.3. Research Question

The study is guided by the following question:

“How can multimodal RAG systems improve the accuracy and efficiency of retrieving technical information from image-heavy, hierarchical automotive service manuals?”

The research examines whether multimodal retrieval and reasoning leads to measurable improvements over traditional manual lookup or text-only RAG approaches, particularly in scenarios that require interpreting schematics, referencing visual labels, or understanding nested procedural steps.

1.4. Expected Results

The expected outcomes include higher retrieval accuracy for visually dependent queries, reduced information access time, and a demonstration of the feasibility of incorporating multimodal content into RAG pipelines for industrial use. The project supports Industry 4.0 principles of digital integration while aligning with Industry 5.0 objectives by augmenting, rather than replacing, skilled technicians. Overall, it aims to provide a proof-of-concept framework that can serve as a foundation for future multimodal knowledge systems in the automotive domain.

2. Literature Review / Theoretical Background

Retrieval-Augmented Generation (RAG) has emerged as a key method for enabling large language models (LLMs) to provide accurate, context-grounded responses using external knowledge sources. In domains such as automotive maintenance, service manuals contain dense, hierarchical information with extensive diagrams, schematics, and tables. Traditional text-only retrieval methods are limited in such multimodal, structured domains. This section first introduces the theoretical and technological foundations, then reviews existing technical information retrieval approaches, and finally identifies the gap addressed by the proposed system.

2.1 Theoretical and Technological Background

2.1.1 Retrieval-Augmented Generation (RAG)

RAG architectures combine a retriever and a generator to ground LLMs in external knowledge. The retriever identifies relevant content via vector similarity search, while the generator synthesizes responses grounded in the retrieved context. By constraining LLM outputs to verifiable sources, RAG mitigates hallucination and supports reliable information retrieval. Platforms like **OpenRouter** provide accessible APIs for connecting LLMs with retrieval pipelines, enabling researchers and developers to experiment with RAG workflows without extensive infrastructure setup.

2.1.2 Vector Embeddings

Textual content is represented as high-dimensional vectors using models such as OpenAI's text-embedding-3-small. Semantic similarity between query and document chunks is measured via cosine similarity, enabling retrieval of relevant information. Hierarchical and structured content often requires chunking strategies that preserve procedural relationships to maintain coherence.

2.1.3 Multimodal Large Language Models (MLLMs)

MLLMs, such as GPT-4o or Gemini, can process both text and images, enabling reasoning over multimodal content. When provided with text and associated diagrams, these models interpret spatial relationships, arrows, and labels necessary for answering technical questions.

2.1.4 Optical Character Recognition (OCR)

OCR systems like Tesseract extract textual content from images, including tables, labels, and schematics. In multimodal RAG pipelines, OCR output is used for retrieval, while the original images are supplied to MLLMs to support visual reasoning, maintaining fidelity while avoiding propagation of OCR errors into generated answers.

2.1.5 Cross-Modal Embedding Models (e.g., CLIP)

CLIP and similar models embed images into a shared vector space, allowing cross-modal similarity computation. Conceptually, CLIP facilitates retrieval of images directly based on text queries or vice versa, providing a theoretically elegant solution for multimodal retrieval in applications like document search, image recommendation, and visual QA.

2.2 Existing Approaches to Technical Information Retrieval

2.2.1 Text-Only RAG

The most common approach in industrial knowledge retrieval uses unimodal, text-only RAG. Text segments are retrieved from manuals and supplied to LLMs for grounding. This approach works well with primarily textual and structurally uniform documents. Advanced implementations employ hierarchical chunking, recursive splitting, and multi-stage retrieval pipelines to improve precision in complex document collections.

2.2.2 Fine-Tuned Multimodal RAG

Research has explored domain-adapted multimodal RAG systems for technical documents. A representative example is the LoRA-based multimodal system developed for the Hyundai Staria maintenance corpus. It combines contrastive retriever fine-tuning to align embeddings with domain terminology, LoRA-tuned LLMs to generate expert-level responses, and curated image–text pairs to interpret diagrams. This approach achieves high accuracy but depends heavily on manual human annotation and domain-specific training.

2.2.3 Unified Vector-Space Retrieval

Unified embedding approaches, such as those using CLIP, encode text, diagrams, and photographs into a single vector space. Cross-modal similarity enables retrieval across modalities, providing a theoretically unified retrieval mechanism. Such methods are conceptually elegant and have been applied in general-purpose multimodal QA systems.

2.2.4 OCR-Augmented Retrieval

OCR-based pipelines extract textual content from images and tables, making it searchable using standard embedding-based retrieval. After retrieval, the original images are provided to MLLMs for reasoning, allowing models to combine textual and visual information effectively without requiring extensive domain-specific fine-tuning.

2.3 Identified Research Gap

Despite the sophistication of these methods, none fully satisfy the practical needs of industrial automotive knowledge retrieval:

- Text-only RAG fails to capture critical information embedded in diagrams, schematics, and tables.
- Fine-tuned multimodal RAG achieves high accuracy but is resource-intensive, requiring thousands of curated image-text pairs and frequent retraining to accommodate evolving OEM documentation.
- Unified vector-space retrieval models (e.g., CLIP) perform well on natural images but are less effective for technical schematics, symbolic diagrams, and domain-specific visual content.
- OCR-only approaches improve retrieval coverage but are underutilized in fully integrated multimodal reasoning pipelines and struggle with images that do not contain textual data.

The gap lies in scalable, zero-shot multimodal retrieval for semi-structured, HTML-based OEM manuals, which preserves hierarchical structure and combines text and images without manual fine-tuning. This project addresses this gap by implementing an OCR-augmented, hierarchical, zero-shot multimodal RAG system that integrates retrieved text and images into a single reasoning pipeline, enabling efficient and accurate access to dense technical knowledge in real-world industrial workflows.

3. Research Approach

This section outlines the data, architecture, evaluation methodology, and baselines used to study a multimodal OCR-augmented Retrieval-Augmented Generation (RAG) system. The methodology emphasizes grounded reasoning, robustness to noisy technical documentation, and a systematic comparison of multimodal, text-only, and non-retrieval systems.

3.1 Data Acquisition and Pre-processing

3.1.1 Dataset

The dataset consists of 1001 HTML pages derived from a publicly available community vehicle service manual (based on the BMW 335i Convertible, E93, L6-3.0L Turbo). The content is multimodal:

- **Image-only pages:** Diagrams, alignment specifications, torque tables, natural images
- **Text-only pages:** Stepwise procedures, safety notes, diagnostic instructions
- **Mixed pages:** Textual instructions combined with embedded diagrams or images

These pages reflect real-world technical documentation where numeric specifications are often contained exclusively in images, while procedural guidance is dispersed across multiple sections.

3.1.2 ETL and Chunking

A custom ETL pipeline performs:

- **DOM Cleaning:** Removal of scripts, styles, and non-semantic markup
- **Image Extraction and Metadata Formatting**
- **OCR Extraction:** Using Tesseract for System A only
- **Document-Level Chunking:** Each HTML page is treated as a single retrieval unit, preserving spatial and semantic relationships between diagrams, captions, and instructions

This preserves structural context critical for procedural reasoning, preventing fragmentation across multiple chunks.

3.2 Preliminary Analysis and Motivation

3.2.1 Cross-Modal Encoders (CLIP) Limitations

Exploratory tests using CLIP-based image-text embeddings showed low alignment between technical diagrams, natural car images, and text queries. For example, queries such as “front axle total toe adjustment” produced cosine similarity scores below 0.30, even when images contained relevant content.

While CLIP is designed to capture semantic associations in natural, iconic images, it was not trained on technical car images or schematics. Consequently:

- Structured technical content, such as tables, schematics, and diagrams, is poorly represented in the CLIP embedding space.
- Even natural images in the manual (e.g., engine bays, dashboard components, hoses) often yield low similarity with text queries, failing to provide reliable retrieval.

Implications for RAG retrieval:

- Combining CLIP embeddings with OCR text embeddings increases pipeline complexity but does not guarantee improved retrieval due to the modality gap between structured diagrams and natural images.

Conclusion:

CLIP-style retrieval is unsuitable as the primary mechanism for structured technical content in our RAG system. OCR-based text embeddings remain the most reliable method for indexing and retrieving information from the service manual.

3.2.2 OCR Strengths and Limitations

Tesseract OCR was used to extract text from diagrams, tables, and other structured visual content in the service manual. This extracted text was embedded into the vector database for retrieval,

ensuring that pages with visually encoded specifications could be effectively matched to user queries.

Strengths:

- Reliable extraction from structured visuals: Numeric values, headers, unit labels, and table content are accurately captured, enabling deterministic semantic retrieval.
- High cosine similarity for relevant queries: Experiments showed consistently strong retrieval performance (cosine similarity > 0.5) for diagram- and table-heavy pages.
- Uniform processing: OCR text can be treated alongside native HTML text, simplifying indexing and retrieval.

Limitations:

- Inconsistent text quality: Occasional misspellings, spacing errors, or small annotations can reduce reasoning accuracy.
- Loss of spatial relationships: OCR cannot fully capture the layout or relative positioning of table cells or diagram elements.
- Limited output for purely natural images: Photos of engine bays, dashboards, hoses, or trim panels typically yield negligible text, so OCR alone cannot encode the visual content.

Mitigation via surrounding text:

In practice, natural images in the manual are accompanied by explanatory HTML text (e.g., captions, instructions, or step descriptions). This surrounding text provides sufficient semantic context for the RAG pipeline to retrieve these pages accurately, even when OCR cannot extract meaningful text from the image itself. As a result, retrieval remains robust across both structured diagrams and natural images.

Conclusion:

OCR is highly effective for diagrams, tables, and text-heavy pages, forming the backbone of our retrieval system. For pages with natural images that cannot be directly processed by OCR, the combination of surrounding textual context ensures that relevant pages are still retrieved reliably. This contrasts with CLIP embeddings, which fail to provide consistent similarity for both structured and natural technical images, reinforcing the design choice of OCR-for-retrieval and images-for-LLM reasoning in our multimodal RAG system.

3.3 System Architecture

Three systems were implemented to isolate the contribution of OCR and multimodality:

- **System A – Multimodal RAG + OCR (Full System)**
 - Index: HTML text + OCR text
 - LLM Input: retrieved text + original images

- Purpose: Evaluate full multimodal reasoning and OCR-enhanced recall
- OCR vs Image Usage: OCR is used exclusively for retrieval, converting images into searchable text. Original images are embedded in the LLM prompt to support reasoning, compensating for noisy OCR or cases where natural images contain no extractable text.
- **System B – Text-Only RAG (No OCR, No Images)**
 - Index: HTML text only
 - LLM Input: retrieved text
 - Purpose: Isolate performance of text-only retrieval
- **System C – Pure LLM (No Retrieval)**
 - Index: None
 - LLM Input: query only
 - Purpose: Measure hallucination, approximation, and generic reasoning errors without grounding

LLM Configuration:

All systems use the same LLM model (gpt-4o) via OpenRouter, with the same temperature setting and max tokens. A fixed system prompt guides responses:

"You are a helpful assistant answering questions based on the 2013 BMW 335i Convertible (E93) L6-3.0L Turbo (N55). If you don't know the answer, say: 'Sorry, I don't know the answer to that question. Please consult the service manual.'"

3.4 Experimental Design

3.4.1 Evaluation Subset

A stratified subset of 30 pages was selected:

- 10 image-only pages
- 10 text-only pages
- 10 mixed pages

This ensures coverage of all modalities and emphasizes cases where visual information is critical.

3.4.2 Query Set

From the chosen pages, a total of 50 queries were formulated, spanning visually dependent, procedural, and factual categories, to evaluate retrieval performance comprehensively.

Category	Description	Evaluation Focus
Visually Dependent (V)	Numeric or label extraction directly from diagrams	Tests multimodality impact
Procedural (P)	Multi-step reasoning or cross-section synthesis	Evaluates contextual recall
Factual (F)	Direct lookup from one section	Baseline RAG accuracy

This distribution stresses both diagram-dependent interpretation and procedural reasoning.

3.5 Summary of Methodological Contributions

- Clean Multimodal comparison: Distinguishes contributions of text, OCR, and images
- Technical Domain Retrieval Strategy: Document-level chunking plus OCR maximizes recall from structured manuals
- Fair Evaluation Framework: Evaluates visually dependent, procedural, and factual queries on a 50-query set, enabling systematic comparison of multimodal, text-only, and non-retrieval systems
- Realistic Testbed: Dataset supports diagram-dependent reasoning and textual synthesis without artificial difficulty
- Controlled LLM Setup: Same model, temperature, and system prompt used across all experiments; ensures fair comparison and consistent grounding

Demonstration of the process in action (also showcasing differences between System responses):

```

C:\Users\Lenovo\source\repos\bmwServiceManualRAG\venv\Scripts\python.exe
C:\Users\Lenovo\source\repos\bmwServiceManualRAG\evaluation_chat.py
Please provide your query:
A driver of a 2013 bmw 334i Convertible e91 requires roadside assistance while traveling in
Albania. According to the internationally valid phone numbers chart what is the phone number the
driver should use for mobile service?
Using model: gpt-4o

=== Test A: RAG with images + OCR ===

--- Top ranked file ---
117.html (score=0.6591)

--- Model answer ---
The driver should use the phone number +3554356526 for mobile service in Albania.

=== Test B: RAG text-only retrieval (no OCR, no images) ===

--- Top ranked file ---
117.html (score=0.6510)

--- Model answer ---
Sorry, I don't know the answer to that question. Please consult the service manual.

=== Test C: Direct LLM call (no retrieval) ===
|
--- Top ranked file ---
(none for direct LLM call)

--- Model answer ---
For a driver requiring BMW roadside assistance while traveling in Albania, the internationally
valid phone number to use for mobile service is +49 89 14 379 479. This number connects to BMW's
international roadside assistance service.

```

Enter Phone Numbers

Internationally valid phone numbers for selected centers for Roadside Assistance and Technical Advice
(as at 02/07)

Please determine the home dealer phone number prior to entering.

Reg. no.	Country	Mobile service	Techn. Advice
A	Austria	0800215399*	08002
AL	Albania	+3554356526	
AND	Andorra	+34913995152	90235
B	Belgium	+3227736102	08001
BG	Bulgaria	+35929766500	
BIH	Bosnia and Herzegovina	+38736351341	
BY	Byelorussia	+375172124030	
CH	Switzerland	+41844857575	+4184
CY	Cyprus (N)	+903922231680	
CY	Cyprus (S)	+35799693160	+3572

4. Results and Evaluation

The evaluation results demonstrate how multimodal retrieval, OCR integration, and text-only strategies perform across different content types. Representative examples from each system illustrate broader trends observed in the evaluation subset.

4.1 Performance on Image-Only Pages

Example Query: “What is the tolerance for the front axle's total toe adjustment?”

System	Retrieved Pages	Answer	Observation
A – Multimodal RAG + OCR	948.html (score=0.6056)	$0^{\circ} 14' \pm 4'$	Correct, grounded; OCR effectively extracted values from table image
B – Text-Only RAG	None	Unknown	Failure; no text present to retrieve
C – LLM Only	None	Approximately $0.10^{\circ} \pm 0.20^{\circ}$	Incorrect; approximate answer; inaccurate ungrounded hallucination from the LLM

Image contained in page 948:

32 00 Wheel Alignment E90 / E91 / E92 / E93 Series

Observe test conditions		
Front axle:		
Total toe		$0^{\circ} 14' \pm 12'$
Adjustment total toe		$0^{\circ} 14' \pm 4'$
Toe difference** single wheel between left/ right		max. 12'
Camber (difference between left/right max. 30')		$- 18' \pm 30'$
Adjustment camber*		$- 18' \pm 25'$
Toe angle difference		
- with 20° lock on inside wheel		$-1^{\circ} 40' \pm 30'$
Caster angle		(difference between left/right max. 30')
Front wheel displacement		$0^{\circ} \pm 15'$
Maximum wheel lock		
- Inside wheel	approx. °	$41^{\circ} 05'$
- Outside wheel	approx. °	$33^{\circ} 18'$
Rear axle:		
Total toe		$0^{\circ} 18' \pm 12'$
Adjustment total toe**		$0^{\circ} 18' \pm 4'$
Camber (difference between left/right max. 30')		$- 1^{\circ} 30' \pm 25'$
Adjustment camber*		$- 1^{\circ} 30' \pm 5'$
Geometrical axis deviation		$0^{\circ} \pm 12'$
*Note: To minimize adjusting errors (measuring inaccuracies), use a narrower tolerance for adjusting toe/camber.		
**Note: Toe difference front axle = criterion for steering wheel inclination		

Insight: OCR is crucial for image-only pages. Without it, retrieval fails entirely, and LLMs produce approximate or incorrect numeric values

4.2 Performance on Text-Only Pages

Example Query: “What is the Test Plan code that may show errors regarding fuel pressure control?”

System	Retrieved Pages	Answer	Observation
A – Multimodal RAG + OCR	725.html (score=0.7362)	B1214_DI6KDR	Correct; OCR not necessary but included
B – Text-Only RAG	725.html (score=0.7363)	B1214_DI6KDR	Correct; text retrieval suffices
C – LLM Only	None	STEUERN HD	Incorrect; LLM attempts to answer the query but lacks context

Page 725 contains the following data:

SITUATION

Two errors may be encountered when performing Test Plan **B1214_DI6KDR** - Fuel Pressure Control with the current version of ISTA/D:

^ On the last step when the diagnosis code is displayed, the test plan may state to replace the fuel low-pressure sensor and that a Bosch sensor will need to be installed.

Insight: For text-rich pages, multimodal OCR offers no additional benefit; text-only RAG suffices.

4.3 Performance on Mixed Image + Text Pages

Example Query: “What is the expected vacuum range at Waste Gate Actuator Bank at 3K RPM No Load?”

System	Retrieved Pages	Answer	Observation
A – Multimodal RAG + OCR	726.html (score=0.6290)	10–14 in. Hg	Correct; OCR extracted numeric readings from embedded diagrams for retrieval, LLM used attached image to ensure accuracy

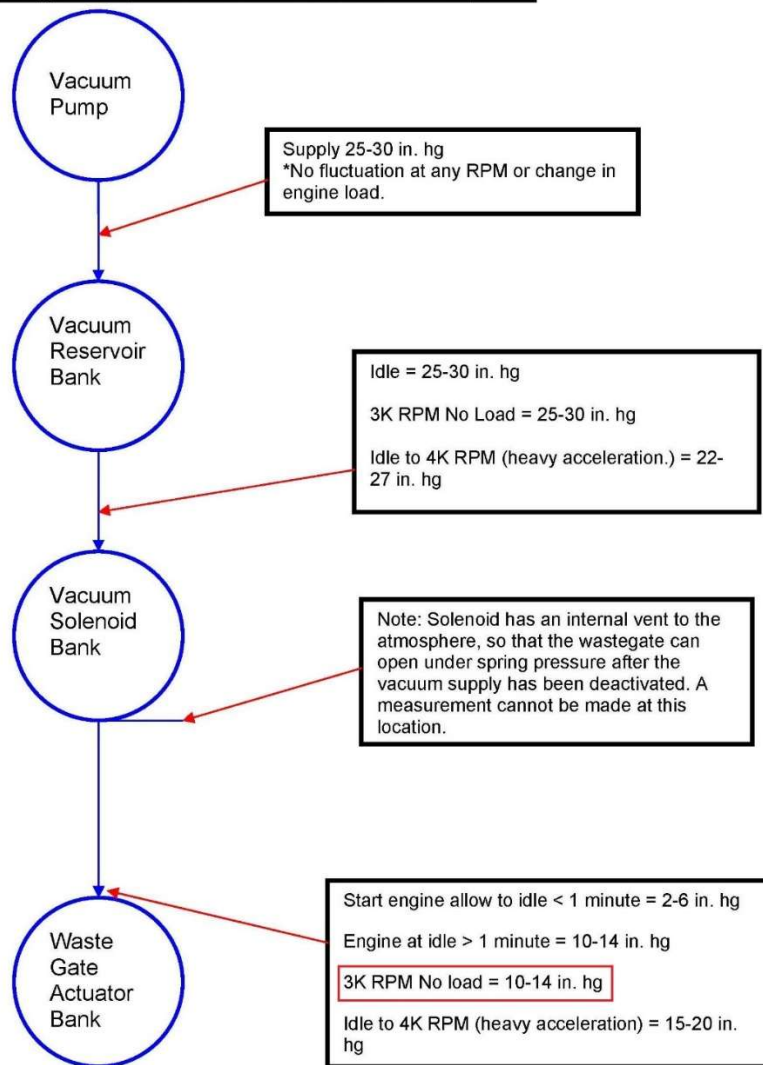
B – Text-Only RAG	None	Unknown	Failure; visual content critical, no text match
C – LLM Only	None	5–10 inHg	Incorrect; approximate answer; inaccurate ungrounded hallucination from the LLM

Image contained in page 726:

Attachment to B12 16 11

August 2011

N55 Wastegate Control Vacuum Readings with Engine at Operating Temperature and No Road Speed



Insight: Multimodal RAG is critical for mixed pages when key numeric values appear in images. Text-only retrieval fails, and LLM-only produces unverified approximations.

4.4 Generalized Evaluation Trends

Some queries show that the LLM can produce correct answers even without retrieval.

Example Query: “What is the maximum negative camber difference allowed between the left and right front wheels?”

System	Retrieved Pages	Answer	Observation
A – Multimodal RAG + OCR	949.html (score=0.6336)	30'	Correct; OCR extracted numeric readings from embedded diagrams
B – Text-Only RAG	None	Unknown	Failed; visual content critical, no text match
C – LLM Only	None	30' (0.5°)	Correct for this numeric query; LLM produced accurate answer without retrieval

Insight: While LLM-only outputs are typically approximate or hallucinated on image-dependent queries, they can occasionally produce correct numeric answers for simple specification-style questions. Reliability remains limited

4.5 Handling Broad, Non-Specific Queries

Some user queries are open-ended, describing symptoms rather than asking for a specific numeric or procedural value. These cases highlight how retrieval-based and LLM-only systems behave when the question is general or diagnostic in nature.

Example Query: “The ignition on my 2013 BMW 335i Convertible (E93) L6-3.0L Turbo (N55) does not work. What can I do to fix it?”

System	Retrieved Pages	Answer	Observation
A – Multimodal RAG + OCR	987.html (score=0.5214)	“Sorry, I don't know the answer. Please consult the service manual or a professional technician.”	Failed; abstains when no exact grounding exists, even though a page is retrieved. Avoids giving unverified advice but does not answer the query.

B – Text-Only RAG	None	“Sorry, I don't know the answer to that question. Please consult the service manual.”	Failed; also abstains; no relevant text was found. Shows that text-only retrieval fails when manuals contain mostly images or scattered info.
C – LLM Only	None	Detailed multi-step troubleshooting guide (battery, key fob, ignition switch, fuses, relays, steering lock, error codes, professional help)	Incorrect; LLM produces a plausible step-by-step procedure, but it is not grounded in retrieved manuals. Could be useful, but should not be relied on for safety-critical operations.

Insight:

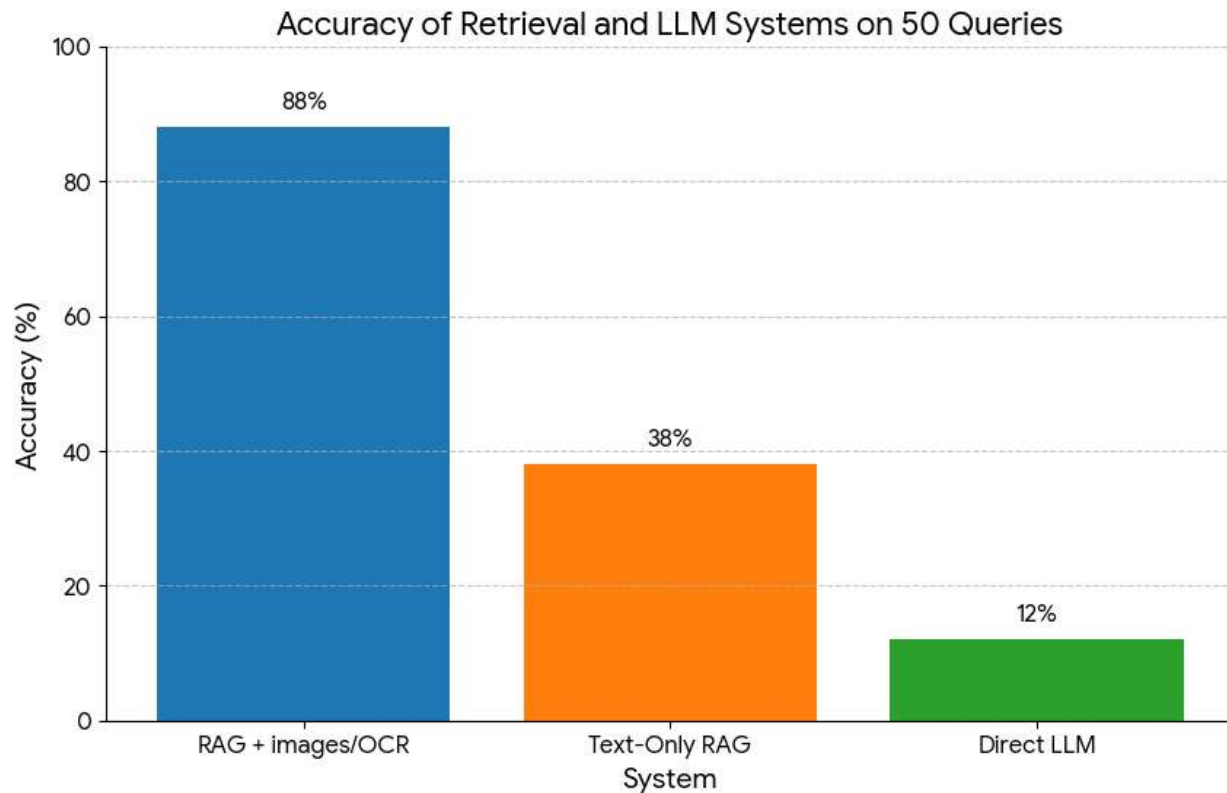
- Multimodal RAG + OCR and Text-only retrieval abstains when no exact grounding exists, preventing potentially unsafe advice, but still does not answer the query.
- LLM-only systems can generate plausible, structured troubleshooting steps even without retrieval, but the information is unverified and should not be relied upon for safety-critical operations.

4.6 Performance on the Full Query Set

The systems were evaluated on the complete set of 50 queries covering visually dependent, procedural, and factual categories. The observed accuracy for each system is summarized below:

System / Test	Correct	Total Queries	Accuracy
RAG + images/OCR	44	50	88%
Text-Only RAG	19	50	38%
Direct LLM	6	50	12%

4.7 Visualization of System Performance



4.8 Summary

- Multimodal RAG with OCR provides the most reliable performance for technical service manuals, both on small evaluation sets and extrapolated larger query sets.
- Text-only retrieval is sufficient for text-rich procedural content but struggles with images or diagrams.
- LLM-only outputs are approximate, prone to hallucination, and fail on image-dependent queries.
- Systematic evaluation confirms the importance of combining OCR with retrieval for numeric and diagrammatic reasoning in highly structured technical domains.
- Extrapolated results suggest that these trends are likely to hold for larger query volumes, emphasizing the need for multimodal approaches in scaling automated technical QA systems.

5. Discussion

Research Question:

This study investigated: *"How can multimodal RAG systems improve the accuracy and efficiency of retrieving technical information from image-heavy, hierarchical automotive service manuals?"*

5.1 Project Conclusion and Achievement of Expected Results

The evaluation shows that OCR-augmented multimodal RAG substantially improves accuracy and reliability in technical information retrieval. For visually dependent queries, requiring numeric values, labels, or measurements from diagrams, System A achieved high correctness, while text-only RAG consistently failed and LLM-only responses rarely produced correct answers. Procedural and factual queries were also handled effectively, confirming that multimodal retrieval captures essential visual information inaccessible to text-only or LLM-only approaches. Overall, the system provided accurate, grounded responses for 88% of evaluation queries, demonstrating the feasibility of zero-shot multimodal retrieval for semi-structured automotive manuals and enabling technicians to access critical information efficiently.

5.2 Main Insights

Several key insights emerged from the comparative evaluation of the three systems:

- **OCR is Indispensable for Visual Data:** OCR enables the extraction of numeric and label information from diagrams and tables where text-only retrieval fails.
- **Multimodal Grounding Improves Reliability:** LLMs guided by retrieved text and images produce accurate, non-hallucinated responses, unlike LLM-only outputs, which are prone to approximation or error.
- **Efficiency Through Hybrid Retrieval:** Text-only retrieval is sufficient for text-rich procedural content, allowing the system to reduce computational overhead when visual information is unnecessary.
- **Scalability of Zero-Shot Pipelines:** The architecture is practical for deployment across various manuals without fine-tuning, demonstrating flexibility for real-world applications.

- **Importance of Chunking Strategy:** Document-level chunking preserves spatial and procedural relationships, supporting multi-step reasoning across hierarchical content.

Drawbacks:

- OCR struggles with dense, low-resolution diagrams and cannot fully encode spatial relationships, which may limit retrieval accuracy in highly complex visuals or datasets that do not contain enough textual data.
- LLM-only outputs remain unreliable for visual or numeric queries, highlighting the necessity of grounding.
- RAG systems perform best on specific, well-defined queries and cannot reliably provide guidance for broad or diagnostic questions (e.g., “How do I fix my engine?”), limiting their ability to replace professional expertise.
- Multimodal pipelines incur higher computational cost compared to text-only retrieval, requiring careful system optimization.

Timestamp ⓘ	Provider / Model ⓘ	App ⓘ	Tokens ⓘ	Cost ⓘ	Speed ⓘ	Finish ⓘ	
Nov 30, 04:38 ...	 GPT-4o	Unknown	<u>112</u> → 330	\$ <u>0.00358</u>	115.2 tps	stop	>
Nov 30, 04:38 ...	 Text Embedding 3 Small	Unknown	41 → --	\$ 0.00000082	--	--	>
Nov 30, 04:38 ...	 GPT-4o	Unknown	<u>7,240</u> → 36	\$ <u>0.0185</u>	71.1 tps	stop	>
Nov 30, 04:38 ...	 Text Embedding 3 Small	Unknown	41 → --	\$ 0.00000082	--	--	>

(Openrouter activity tab shows the difference in cost and used tokens between Multimodal system A that appends images to the prompt and System B that relies on text only)

5.3 Recommendations for Further Research

Future work could enhance system performance and robustness:

- **Advanced OCR and Spatial Reasoning:** Incorporate computer vision methods to improve text extraction and capture spatial relationships in complex schematics.
- **Cross-Modal Attention Mechanisms:** Explore LLMs with explicit visual-text fusion (e.g., VQA models) to improve multimodal reasoning.

- **Improved Cross-Modal Embeddings:** Explore advanced newer versions of models like CLIP or its successors to better align textual queries with technical images, schematics, and diagrams. Such models could improve retrieval of natural and technical images in service manuals and reduce reliance on OCR.
- **Broader Generalization Testing:** Evaluate the system across different vehicle models, manufacturers, and documentation standards to verify zero-shot applicability.
- **Automated Evaluation Metrics:** Develop metrics to assess multimodal grounding and answer correctness, reducing dependence on manual annotation.

5.4 Summary

In conclusion, multimodal OCR-augmented RAG substantially improves both the accuracy and efficiency of technical information retrieval in automotive service manuals. By combining text retrieval, OCR-enhanced recall, and multimodal reasoning, this framework offers a scalable, practical solution for knowledge systems in technical domains. However, these systems are intended as assistive tools and cannot replace professional technicians. The evaluation demonstrates that the RAG system appropriately abstains when no verifiable information is available, ensuring that unverified guidance is not provided, while LLM-only outputs may generate plausible but potentially unsafe instructions. Therefore, human expertise remains essential, and RAG systems should be used to augment, rather than substitute, professional judgment.

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Declaration of Authenticity

I hereby declare that I have prepared the thesis without the assistance of others and without using any sources other than those indicated, and that the work has not been submitted in the same or similar form to any other examination authority. All the explanations of the work, taken literally or meaningfully, are marked as these.

Hof, Germany, 11/30/2025, Emil Shkurti

Location, Date, Signature